

Algebra II with Trigonometry

Chapter 5.4

Olympiad Solutions (Gemini)

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Solutions

1. Let $\theta_n = \tan^{-1} \left(\frac{1}{\sqrt{n}} \right)$. Evaluate the exact value of the product:

$$\cos(\theta_1) \cdot \cos(\theta_2) \cdot \cos(\theta_3) \cdots \cos(\theta_{99})$$

Solution: If $\theta_n = \tan^{-1} \left(\frac{1}{\sqrt{n}} \right)$, we can visualize a right triangle with an opposite side of length 1 and an adjacent side of length $\sqrt{n}$. By the Pythagorean theorem, the hypotenuse is $\sqrt{1^2 + (\sqrt{n})^2} = \sqrt{n+1}$. Thus, the cosine of the angle is given by:

$$\cos(\theta_n) = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{\sqrt{n}}{\sqrt{n+1}}$$

The given expression transforms into a telescoping product:

$$\prod_{n=1}^{99} \cos(\theta_n) = \frac{\sqrt{1}}{\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{3}} \cdot \frac{\sqrt{3}}{\sqrt{4}} \cdots \frac{\sqrt{99}}{\sqrt{100}}$$

Canceling all the intermediate numerator-denominator pairs leaves only the first numerator and the final denominator:

$$\frac{\sqrt{1}}{\sqrt{100}} = \frac{1}{10}$$

2. Evaluate the exact value of the expression below (where the angle 5 is measured in radians):

$$\sin^{-1}(\sin 5) + \cos^{-1}(\cos 5)$$

Solution: Because 5 radians falls in the fourth quadrant (since $\frac{3\pi}{2} \approx 4.71 < 5 < 2\pi \approx 6.28$), we can find an acute reference angle $\alpha = 2\pi - 5$ which lies in the interval $(0, \frac{\pi}{2})$. In the fourth quadrant, sine is negative and cosine is positive. Therefore:

$$\sin 5 = -\sin(2\pi - 5) = -\sin \alpha$$

$$\cos 5 = \cos(2\pi - 5) = \cos \alpha$$

Applying the inverse functions, which evaluate principal values:

$$\sin^{-1}(\sin 5) = \sin^{-1}(-\sin \alpha) = -\alpha = 5 - 2\pi$$

$$\cos^{-1}(\cos 5) = \cos^{-1}(\cos \alpha) = \alpha = 2\pi - 5$$

Adding the two results:

$$(5 - 2\pi) + (2\pi - 5) = 0$$

3. Evaluate the exact value of the following nested composition, expressing your final answer in radians:

$$\cos^{-1} \left(\sin \left(\tan^{-1} \left(\frac{1}{\sqrt{3}} \right) + \tan^{-1}(1) \right) \right)$$

Solution: First, evaluate the standard inverse tangents:

$$\tan^{-1} \left(\frac{1}{\sqrt{3}} \right) = \frac{\pi}{6} \quad \text{and} \quad \tan^{-1}(1) = \frac{\pi}{4}$$

The expression inside the sine function becomes their sum:

$$\frac{\pi}{6} + \frac{\pi}{4} = \frac{5\pi}{12}$$

We now need to evaluate $\cos^{-1} \left(\sin \left(\frac{5\pi}{12} \right) \right)$. Utilizing the elegant cofunction identity $\sin x = \cos \left(\frac{\pi}{2} - x \right)$:

$$\sin \left(\frac{5\pi}{12} \right) = \cos \left(\frac{\pi}{2} - \frac{5\pi}{12} \right) = \cos \left(\frac{\pi}{12} \right)$$

Because $\frac{\pi}{12}$ falls strictly within the principal domain of the inverse cosine function $[0, \pi]$, we have:

$$\cos^{-1} \left(\cos \left(\frac{\pi}{12} \right) \right) = \frac{\pi}{12}$$

4. Using the reciprocal properties of inverse trigonometric functions, evaluate the exact value of:

$$\sin \left(\sec^{-1} \left(\frac{5}{4} \right) + \csc^{-1} \left(\frac{5}{3} \right) \right)$$

Solution: Applying the reciprocal properties $\sec^{-1}(x) = \cos^{-1} \left(\frac{1}{x} \right)$ and $\csc^{-1}(x) = \sin^{-1} \left(\frac{1}{x} \right)$, the expression becomes:

$$\sin \left(\cos^{-1} \left(\frac{4}{5} \right) + \sin^{-1} \left(\frac{3}{5} \right) \right)$$

Notice that both angles describe the same standard 3-4-5 right triangle. Let $\alpha = \sin^{-1} \left(\frac{3}{5} \right)$. This implies $\sin \alpha = \frac{3}{5}$ and $\cos \alpha = \frac{4}{5}$, which means $\cos^{-1} \left(\frac{4}{5} \right)$ is also equal to α . The expression neatly reduces to:

$$\sin(\alpha + \alpha) = \sin(2\alpha)$$

Using the sine double-angle formula:

$$\sin(2\alpha) = 2 \sin \alpha \cos \alpha = 2 \left(\frac{3}{5} \right) \left(\frac{4}{5} \right) = \frac{24}{25}$$

5. Find the exact positive real number x that satisfies the equation:

$$\sin^{-1}(x) + \sin^{-1}(2x) = \frac{\pi}{2}$$

Solution: Rearrange the equation to isolate $\sin^{-1}(2x)$:

$$\sin^{-1}(2x) = \frac{\pi}{2} - \sin^{-1}(x)$$

Using the complementary identity $\frac{\pi}{2} - \sin^{-1}(x) = \cos^{-1}(x)$, the equation elegantly simplifies to:

$$\sin^{-1}(2x) = \cos^{-1}(x)$$

Take the sine of both sides. By the right triangle equivalence $\sin(\cos^{-1}(x)) = \sqrt{1 - x^2}$ for $x > 0$:

$$2x = \sqrt{1 - x^2}$$

Square both sides to solve for x :

$$4x^2 = 1 - x^2 \implies 5x^2 = 1$$

Since we are given that x is positive, we take the positive square root:

$$x = \frac{1}{\sqrt{5}} = \frac{\sqrt{5}}{5}$$

6. In a right triangle with a hypotenuse of length 10, the two acute angles α and β satisfy the equation:

$$\tan^{-1}(\sin \alpha) + \tan^{-1}(\cos \beta) = \tan^{-1}\left(\frac{8}{15}\right)$$

Find the exact length of the triangle's shortest leg.

Solution: In a right triangle, the acute angles α and β are complementary ($\alpha + \beta = \frac{\pi}{2}$). This intrinsic cofunction relationship dictates that $\cos \beta = \sin \alpha$. Let $x = \sin \alpha$. The given equation simplifies directly to:

$$2 \tan^{-1}(x) = \tan^{-1} \left(\frac{8}{15} \right)$$

Take the tangent of both sides and apply the tangent double-angle formula, $\tan(2\theta) = \frac{2 \tan \theta}{1 - \tan^2 \theta}$:

$$\frac{2x}{1 - x^2} = \frac{8}{15}$$

Cross-multiplying gives $30x = 8 - 8x^2$, which rearranges into the quadratic equation:

$$4x^2 + 15x - 4 = 0$$

Factoring the quadratic reveals $(4x - 1)(x + 4) = 0$. Since $x = \sin \alpha$ must be strictly positive, $x = \frac{1}{4}$. The shortest leg of the right triangle is opposite its smallest angle (which is α , since $\sin \alpha < \sin \beta$). Its length is strictly defined by the hypotenuse multiplied by $\sin \alpha$:

$$10 \sin \alpha = 10 \left(\frac{1}{4} \right) = \frac{5}{2}$$